

Boston Food Safety Inspections Trend (2005-2026): Volume, Severity, and Neighborhood Change

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Background

Boston's health inspections spans 20+ years of restaurant data. The public data includes outcomes on inspections, pass or fails and violation levels, which can be good to reveal public health trends and neighborhood disparities.

Research Question

To what extent have Boston **food safety inspection outcomes** changed over **time** in terms of **volume** and **severity**, and are certain **neighborhoods** improving or worsening over the past 20 years?

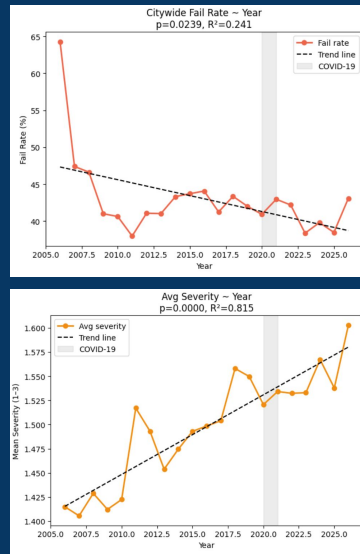
Data & Method

- **Source:** Analyze Boston open data portal – Boston Inspectional Services Division food inspection records, 2005-2026.
- **Unit of observation:** Raw data has one row per violation. Rows were grouped by business name and inspection date to reconstruct one row per inspection, with a new column counting violations per inspection.
- **Outcome variable:** Result column standardized and merged to only contain "Pass" and "Fail" All other outcomes dropped.
- **Severity:** Violation level (* / ** / ***) converted to numeric (1 / 2 / 3) to enable quantitative analysis. Missing severity (rows with no violation, i.e. passing inspections) treated as NaN and excluded from severity calculations but kept in fail rate calculations.
- **Neighborhood:** City/neighborhood labels cleaned and unified. Neighborhoods with fewer than 200 inspections total excluded as estimates were too noisy. The generic "BOSTON" label (~74k rows) excluded as it is a catch-all, not a true neighborhood.
- **Time:** Inspection date parsed to extract year. Rows with unparseable dates dropped. Analysis spans 2005-2026.

Limitations

- Small neighborhoods → noisy estimates
- COVID-19 distorts 2020-2021 as we see inspection volumes drop
- Confounders to consider: enforcement bias, language barriers, inspector behavior

Citywide trends over time



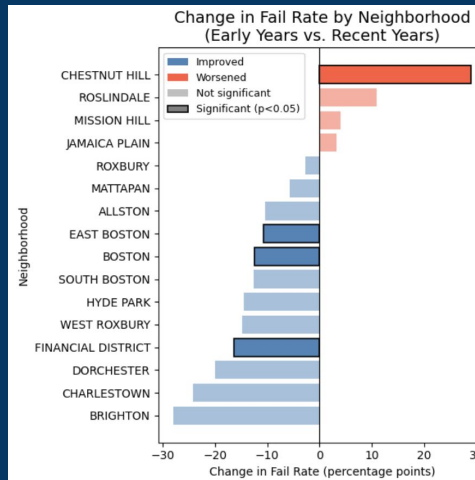
Fail Rate declines (slope = -0.0043 per year) while Average Violation Severity trends upwards (slope = 0.00824 per year). Over the years, there are fewer violations overall but more serious ones.

Regression Model

	Fail Rate ~ Year	Severity ~ Year
Slope	-0.00430	0.00824
R ²	0.241	0.815
p-value	0.0239	> 0.001

The **severity model fits well** ($R^2 = 0.815$), consistent with a linear trend over time. The **fail rate model fits more poorly** ($R^2 = 0.241$), likely because the early years (2005-2007) show unusually high fail rates that act as outliers and pull the trend line. The relationship may not be strictly linear, which we can assume from the more plateaued data from 2008 forward, and the 2026 underprediction.

Neighborhood change scores



Neighborhood level regression shows 3 neighborhoods with statistically significant declining trends (Brighton, Charlestown, Dorchester) and 1 with significant worsening (Chestnut Hill).

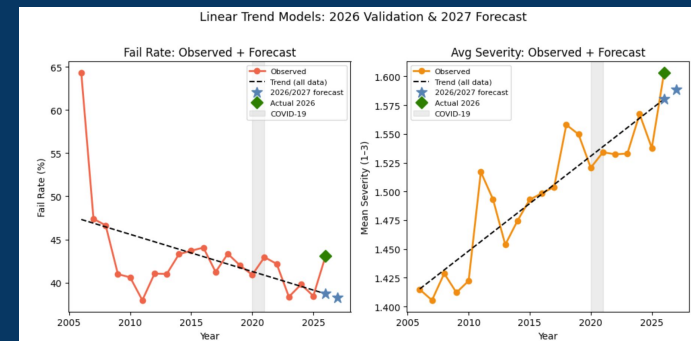
Most Boston neighborhoods saw a decline in fail rates over the past 20 years, with Brighton, Charlestown, and Dorchester showing the largest improvements. Gray neighborhoods had insufficient inspection data for reliable estimates

Choropleth map (change in fail rate)



2026 Prediction & 2027 Forecast

Using 2005-2025, the linear model predicted a 2026 citywide fail rate of 37.77%, with the actual 2026 fail rate being 43.07% (underestimate of 5.30). For severity, predicted 1.575 but the actual was 1.603 (underestimate of 0.028). Both metrics were **underpredicted** in 2026. Projecting forward, the model forecasts a 2027 fail rate of 38.28% and an average severity of 1.59, though given the 2026 under predictions these should be interpreted as **conservative estimates**.



Conclusion

Boston's citywide **fail rate declined** significantly over 20 years (slope = -0.0043/yr, $p = 0.024$), while **average violation severity increased** significantly (slope = +0.0082/yr, $p < 0.001$) – suggesting fewer violations overall, but more serious ones. At the neighborhood level, **3 neighborhoods improved significantly** and **1 worsened**. The model underestimated the 2026 fail rate by 5.30 pp, pointing to a likely post-COVID rebound not captured by a linear trend. Limitations include the "BOSTON" catch-all label, small neighborhood sample sizes, and potential enforcement and reporting biases.

Reference

City of Boston Inspectional Services Department. (2026). Food establishment inspections [Dataset]. Analyze Boston / Department of Innovation and Technology. <<https://data.boston.gov/dataset/food-estab-lishment-inspections>>